AI-Based Adaptive VR Meditation System

INDIVIDUAL THESIS

Prathikesh Senthilkumaran

IT22621788

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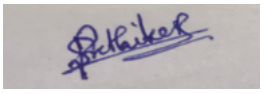
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Student name: S. Prathikesh

Student ID: IT22621788

Signature: _____  _____

Date: ___11/10/2026_____

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Role and name	Signature	Date
Supervisor: Mr. Samadhi Rathnayake		11/10/2026
Co-supervisor: Ms. Malithi Nawarathne		

ABSTRACT

Adaptive virtual-reality meditation systems need a dependable, non-invasive way to tell whether a session has reduced a user stress, and the voice is often the only affective signal available without additional wearables. Voice-based stress detectors increasingly reuse frozen, emotion-pretrained speech representations, conventionally selected by how well they agree with acted emotional speech in-domain. This thesis develops the voice-based stress assessment and cross-modal session-validation component of such a system and reports a diagnostic finding: on real, spontaneous stress this selection convention is not merely uninformative but inverted. Across six fusion checkpoints spanning acted-to-natural training, higher in-domain arousal agreement coincides with lower real-voice accuracy (Spearman rho = -0.60), the checkpoint the convention would reject (0.35 in-domain) deploys best, at 91.7% leave-one-out accuracy.

The cause is a split in what frozen emotion embeddings preserve: they retain the valence of real stress but lose its arousal. On held-out real clips, stressed speech is correctly negative in valence 82-100% of the time, yet correctly high in arousal only 12-53% of the time. The same asymmetry replicates on the independent IEMOCAP corpus (n = 2,132), and a matched-size probe (emotion2vec versus WavLM) shows the valence advantage is specific to emotion pretraining rather than model capacity. Acting on the diagnosis, a valence-primary scoring rule widens stressed/calm separation over the conventional arousal-based rule on every checkpoint (Cohen's d = 1.71).

The finding is deployed as a bilingual detector, built on a frozen emotion2vec encoder with a trainable prosodic branch, that defers arousal magnitude to a physiological heart-rate-variability module by design. It sits within a five-layer pipeline spanning ambient checking, acoustic stress scoring, pre/post change analysis, cross-modal validation, and autoencoder-based session anomaly detection. The individual contribution is this diagnosis together with the scoring rules and system that follow it.

Keywords: voice-based stress detection, speech emotion recognition, valence and arousal, frozen embeddings, multilingual affective computing

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
VR	Virtual Reality
SER	Speech Emotion Recognition
SSL	Self-Supervised Learning
CCC	Concordance Correlation Coefficient
HRV	Heart Rate Variability
LOO	Leave-One-Out
VAD	Voice Activity Detection
STT	Speech-to-Text
TTS	Text-to-Speech
API	Application Programming Interface
GeMAPS	Geneva Minimalistic Acoustic Parameter Set
RMS	Root Mean Square
dBFS	Decibels relative to Full Scale
OOD	Out-of-Distribution
DSR	Design Science Research

1 INTRODUCTION

1.1 Overall project context

Stress is one of the most pervasive challenges to mental health and quality of life, and access to professional stress-relief support remains limited for many people because of cost, availability, and stigma [1], [2]. Virtual-reality (VR) meditation is a promising and accessible response, and controlled studies report that immersive VR relaxation and heart-rate-variability biofeedback can measurably reduce perceived stress [4], [5].

This thesis forms one component of the AI-Based Adaptive VR Meditation System (project R26-DS-006), a multimodal, stress-aware platform that adapts a meditation experience to the user in real time. As shown in Figure 1, the platform is organized into four cooperating components. Component A is an adaptive audio controller, Component B performs multi-level physiological stress inference from heart-rate-variability (HRV) features, Component C personalizes the VR environment, and Component D, the subject of this thesis, performs voice-based stress assessment and session-level outcome validation.

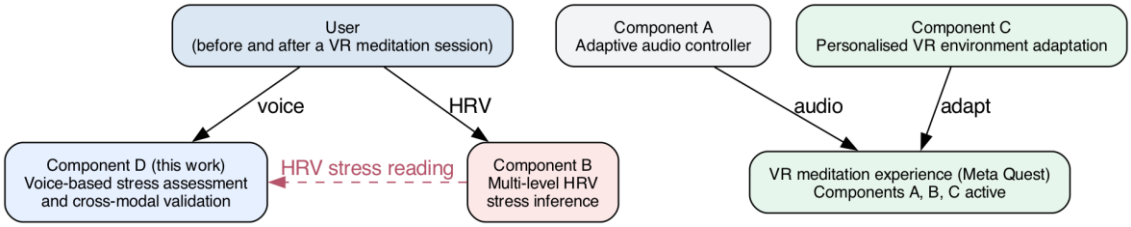


Figure 1: Overall architecture of the AI-Based Adaptive VR Meditation System and the place of Component D within it.

Together, these components form a closed adaptive loop. Physiological and behavioural signals estimate the user's stress, the audio and environment adapt accordingly, and the session outcome is evaluated so the platform can report, and ultimately learn from, whether the intervention helped. Component D supplies the behavioural (voice) signal and the per-session verdict, and cross-validates its reading against Component B's physiological estimate so that neither modality is trusted blindly.

Unmanaged stress is linked to reduced cognitive performance, emotional difficulty, and long-term physical health consequences, and the barriers to professional support, namely cost, waiting times, availability, and stigma, leave many people without timely help. This motivates accessible, technology-mediated interventions that a person can use at home without specialist supervision, and virtual-reality meditation is one of the more promising of these because it is immersive, engaging, and increasingly affordable on consumer headsets.

What has been missing from such systems is a dependable way to confirm, for an individual session, that the intervention helped. Existing platforms either assume benefit or rely on the user by filling in a questionnaire afterwards, which is subjective and easily skipped. The AI-

Based Adaptive VR Meditation System addresses this by closing the loop with real signals: physiology from a wearable, behaviors from the voice, and adaptation of the audio and environment, so that the experience is tailored to the user and its effect is measured rather than assumed.

1.2 Individual component context

The system developed by the author is Component D, the voice-based stress assessment and cross-modal session-validation layer. It answers a focused question for the wider platform: did the VR meditation session actually reduce this user's stress, measured non-invasively from the user's own voice, and across more than one language? The component records a short voice sample before the session and again after it, scores acoustic stress from each, compares the two, cross-validates the voice reading against Component B's HRV-based stress, and flags sessions that deviate from the normal stress-reduction pattern.

A deliberate design decision is that stress is scored from how the user speaks, through acoustic and prosodic characteristics of the signal, rather than from what is said. Because the score reads the raw audio and not a transcript, the same scoring path applies across languages. The current implementation targets English and Sinhala, with Tamil as a planned extension, and the component is realised as a five-layer pipeline described in detail in Chapters 4 and 5.

The user never operates a record button. A companion persona guides the check-in as a short, natural conversation, recording is automatic in the background, and the companion speaks one line at a time, so the interaction feels unhurried. This design is deliberate: performance anxiety and fiddly controls change how people speak, so removing them yields more natural speech and therefore a more honest acoustic reading. The same guided flow runs before the VR session and again afterwards, bracketing the experience so that the within-speaker change can be measured.

1.3 Problem statement

Reading stress from the voice is attractive because it is passive, inexpensive, and needs only a microphone. In a session-adaptive well-being system, it is often the only affective signal available without additional wearables. The practical difficulty is not whether stress leaves a vocal trace, but whether it can be read reliably from a few seconds of real, spontaneous speech, spoken by an unseen speaker, in more than one language, and without first gathering a large labelled in-domain corpus. Three issues make this hard.

1. Reliance on acted data. The standard way to handle the data problem is to reuse a frozen emotion-pretrained encoder and train only a small head on top. Such models are conventionally selected by how well they agree with acted emotional speech, yet acted corpora skew toward loud, high-arousal portrayals and may not represent real stress.

2. Quiet, internalized stress. Real stress, especially in students and desk workers, often appears as a quiet, flattened response that is negative in tone but low in activation. If what a frozen encoder carries over from acted data is not what separates real stress, selecting models by in-domain agreement may be misleading.

3. No automated session validation. Existing VR meditation platforms verify effectiveness through self-report questionnaires or laboratory equipment, and provide no automated, voice-based mechanism to confirm that a given session reduced the user's stress, nor any way to reconcile a voice reading with a physiological one.

1.4 Research gap

A review of recent work shows that although voice-based stress detection, multimodal stress recognition, and VR stress interventions are each individually mature, their synthesis into a multilingual, cross-modal, self-validating voice component is not addressed, and the selection of frozen emotion encoders for real-voice deployment has not been tested directly. Table 1 compares the capabilities of representative prior to work against this thesis.

Capability	**EMOVOME [18]**	**WESAD [25]**	**VR platforms (TRIPP) [4]**	**This work**
Voice-based session effectiveness evaluation	No	No	No	Yes
Cross-modal voice and HRV validation	No	Partial	No	Yes
Multilingual (English and Sinhala)	No	No	No	Yes
Evaluated on real (not acted) stress voices	Yes	Partial	No	Yes
Valence versus arousal transfer analysis	No	No	No	Yes
Frozen-encoder selection diagnosis	No	No	No	Yes
Session anomaly detection	No	No	No	Yes

Table 1: Research gap analysis: capabilities of representative prior work against this thesis.

The primary gap this thesis addresses is empirical and diagnostic. It has not been established, for real-voice deployment, whether ranking frozen emotion embeddings by in-domain acted agreement predicts real-voice stress accuracy, nor which affective axis, valence or arousal, survives the shift from acted training data to real spontaneous stress. A secondary gap is systemic: no existing VR meditation system measures stress from the voice before and after a session, cross-validates it against HRV, and detects unusual sessions within a single multilingual component.

1.5 Aim and objectives

1.5.1 Main aim

To design, implement, and evaluate a multilingual voice-based stress assessment and cross-modal session-validation component for an AI-based adaptive VR meditation system, and in doing so to establish which affective axis a frozen emotion encoder transfers to real spontaneous stress and to derive the scoring rule that follows from that finding.

1.5.2 Specific objectives

1. Develop a pre-session ambient-environment assessment that decides whether the user's acoustic environment is suitable for reliable voice capture.
2. Build a voice-based acoustic stress-scoring model on a frozen emotion-pretrained encoder with a lightweight trainable head and establish through an ablation which affective axis transfers to real voices.
3. Test whether an in-domain acted agreement predicts real-voice stress accuracy and replicates the resulting finding at scale on an independent corpus.
4. Derive and validate a scoring rule consistent with the finding and confirm it improves the separation between stressed and calm speech.
5. Implement pre- and post-stress-change analysis and a cross-modal validation framework against Component B's HRV-based stress.
6. Design and train an autoencoder-based session anomaly detector on session-level stress features.
7. Deliver the component as a bilingual English and Sinhala system that meets at least 75 percent stress-classification accuracy across both languages on held-out real data and keep the design extensible to further languages.

1.6 Research questions or hypotheses

1.6.1 Research questions

RQ1. Does in-domain acted agreement predict real-voice stress accuracy?

RQ2. Which affective axis, valence or arousal, does a frozen emotion encoder transfer to real stress?

RQ3. Is any valence-over-arousal asymmetry specific to emotion pretraining rather than model size?

RQ4. Does a valence-primary scoring rule improve the separation between stressed and calm speech?

RQ5. How does the resulting detector behave across an in-distribution language (English) and an out-of-distribution one (Sinhala)?

1.6.2 Research hypotheses

H1. Selecting frozen emotion encoders by in-domain acted agreement is not predictive of real-voice accuracy, and the relationship is in fact inverted.

H2. Frozen emotion embeddings transfer valence reliably but not arousal, so a valence-primary rule improves separation on every checkpoint.

1.7 Scope limitations and assumptions

1.7.1 Scope

The component covers English and Sinhala episodic (before and after) stress assessment, delivered as an Android mobile application communicating with a backend service that hosts the models. The acoustic model uses a frozen emotion encoder with a small trainable head, and arousal magnitude is handed to Component B's HRV module by design. Tamil is in scope at the interface level and is a planned scoring extension.

1.7.2 Limitations

The real-voice evaluation set is small (24 clips, roughly three Sinhala speakers), so per-language estimates are reported with bootstrap intervals, and large-sample inference is carried by an independent corpus. Arousal magnitude is unreliable by design and is offloaded rather than repaired, and a single absolute score for an unseen voice is less reliable than the within-speaker change.

1.7.3 Assumptions

Recordings are short, the speaker is cooperative, and the environment is quiet enough to pass the Layer 1 gate. Component B's physiological estimate is assumed to be available for cross-modal validation when a wearable is connected, communicated through a standardized schema.

1.8 Individual contribution

The author designed, built, and evaluated Component D, which forms Chapters 4, 5, and 6 of this thesis. The specific individual contributions are:

1. A demonstration that ranking frozen emotion embeddings by in-domain acted agreement is anti-correlated with real-voice stress accuracy (Spearman rho of -0.60), reversing a common selection habit.

2. An explanation of the cause as a valence and arousal split, where frozen emotion embeddings carry the valence of real stress but not its arousal, replicated at scale on IEMOCAP (2,132 samples) with bootstrap significance.

3. Evidence that the split is specific to emotion pretraining rather than model size, using a matched-size probe comparing emotion2vec and WavLM.

4. A valence-primary scoring rule that follows the finding and widens the separation between stressed and calm speech on every checkpoint (Cohen's d of 1.71).

5. An engineered five-layer, bilingual voice-stress component with confidence-gated cross-modal validation, companion-guided capture, and graceful degradation, reaching 91.7 percent leave-one-out accuracy.

1.9 Thesis structure

The remainder of this thesis is structured as follows. Chapter 2 reviews voice-based stress detection, dimensional emotion and its evaluation, multimodal and cross-modal stress recognition, and multilingual speech emotion recognition, and synthesizes the gap into a conceptual framework. Chapter 3 presents the research methodology, including the design, data, procedure, metrics, analysis, ethics, and reproducibility. Chapter 4 describes the requirements and detailed design of the component. Chapter 5 covers implementation, integration, testing, and deployment. Chapter 6 reports the research and product results and their validation. Chapter 7 interprets the findings, compares them with the literature, and states contributions, limitations, and implications. Chapter 8 concludes and sets out future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This review combines two bodies of literature that the component bridges: the applied literature on stress, well-being, and VR interventions, and the technical literature on speech-based affective computing and multimodal stress recognition. Sources were gathered from IEEE Xplore, the ACL Anthology, PubMed and PLOS, Scopus, and, for self-supervised speech encoders where the canonical references are recent, arXiv and conference proceedings, covering publications from 2018 to 2026.

Search terms combined the component's core concepts: voice-based stress detection, speech emotion recognition, valence and arousal, frozen or self-supervised speech representations, heart-rate-variability stress, multimodal and cross-modal stress recognition, low-resource and multilingual speech emotion recognition, Sinhala speech, autoencoder anomaly detection, and VR meditation or biofeedback. Peer-reviewed studies were prioritised, with a small number of widely cited preprints retained where they are the standard reference for a method. The review concludes by synthesising the identified gaps into the conceptual framework that structures the methodology in Chapter 3.

Studies were included when they reported empirical results on stress, affect, or emotion from speech or physiological signals, used reproducible methods, and were relevant to real-world or multilingual deployment. Purely theoretical pieces, non-peer-reviewed sources without a standard-reference status, and studies lacking any quantitative evaluation were excluded. Where a topic was covered by many works, the most cited and the most methodologically rigorous were retained, so the review foregrounds depth on the question of which affective information a frozen encoder transfers rather than breadth across every emotion-recognition result.

2.2 Domain and theoretical foundations

2.2.1 Stress, well-being, and VR meditation

Stress-related conditions are a leading contributor to reduced quality of life, and access to professional support is constrained by cost, availability, and stigma [1], [2], [3]. VR meditation and biofeedback are a credible consumer-scale response, with randomised and controlled studies reporting measurable reductions in perceived stress and improvements in physiological indicators, including HRV biofeedback delivered through head-mounted displays [4], [5]. These findings establish both the value of the intervention and the need for a dependable way to verify, per session, whether it worked.

The economic and social cost of unmanaged stress is substantial, and consumer interest in digital wellness tools has grown accordingly. Controlled evaluations of VR relaxation report reductions in self-reported stress and improvements in physiological markers such as heart-rate variability, and head-mounted delivery tends to outperform screen-based delivery because immersion reduces external distraction. These results justify the intervention but also expose

the evaluation gap that this component fills, since they rely on instruments and laboratory conditions that a home user does not have.

2.2.2 Voice as a stress signal and dimensional emotion

Acoustic and prosodic properties of speech respond to physiological stress. Fundamental frequency, jitter, shimmer, and speaking rate shift under psychosocial stressors, and importantly track physiological stress rather than isolated negative affect [6], [7]. The Geneva Minimalistic Acoustic Parameter Set codifies a compact, interpretable feature set for this purpose [17], and informs the prosodic front-end used here. Most modern affect models place emotion on the two-dimensional valence and arousal plane of the circumplex model [12], where valence captures how positive or negative a state is and arousal how activated or calm. Continuous recognition on these axes is conventionally scored with the concordance correlation coefficient (CCC) [13], on acted benchmarks such as IEMOCAP [14] and MELD [15].

Several acoustic correlates recur across the stress of literature. Fundamental frequency and its variability rise under acute stress as laryngeal muscle tension increases; jitter and shimmer, which measure cycle-to-cycle instability in pitch and amplitude, grow as phonation becomes less regular; the harmonics-to-noise ratio falls as the voice becomes breathier or rougher; and timing measures such as speaking rate and pause ratio shift as cognitive load changes. These are the measures the prosodic branch of this component computes, so that the model has access to interpretable stress-sensitive cues alongside the learned embedding.

The concordance correlation coefficient used to score valence, and arousal regression combines the Pearson correlation between predictions and targets with a penalty for any difference in their means and variances. A model can therefore achieve a high correlation, yet a low concordance if its predictions are systematically shifted or compressed, which makes the coefficient a stricter and more appropriate measure than correlation alone for dimensional affect.

2.2.3 Frozen self-supervised speech representations

Self-supervised encoders learn general-purpose features from large unlabeled corpora and are now the default front-end for speech tasks. wav2vec 2.0 [9] and WavLM [10] optimize masked-prediction or contrastive objectives and are commonly used with a frozen backbone under a small trainable probe, so only a few hundred thousand parameters need training. emotion2vec [11] adapts this recipe to affect by pretraining emotional speech. Freezing the encoder is parameter-efficient and stable, but it fixes whatever the pretraining captured, so the deployment question is which affective information survived pretraining, which is exactly what the shift from acted to real stress puts to the test.

The three encoders compared in this thesis differ in what their pretraining emphasizes, summarized in Table 3. wav2vec 2.0 and WavLM are trained on large general-speech corpora with masked-prediction or denoising objectives and are optimized for the full stack of speech tasks, so their representations foreground phonetic and speaker content. emotion2vec is pretrained on emotional speech with an affect-oriented objective, so its representation

foregrounds emotional content. This difference is the reason, developed in Chapter 6, that emotion2vec transfers valence to real stress while a full-stack encoder does not.

Encoder	Pretraining emphasis	Representation foregrounds
wav2vec 2.0	General speech, contrastive masked prediction	Phonetic and acoustic content
WavLM	Large-scale speech, masked speech denoising and prediction	Full-stack speech, including speaker
emotion2vec	Emotional speech, affect-oriented self-supervision	Emotional (affective) content

Table 2: Self-supervised speech encoders compared to this thesis and what their pretraining emphasizes.

2.3 Related research

2.3.1 Voice-based stress detection

Kappen et al. establish that acoustic and prosodic features reflect physiological stress specifically, positioning voice as an objective correlate rather than a purely subjective cue [6], while a systematic review confirms the prominence of prosodic markers and their sensitivity to recording conditions [7]. A meta-analysis of fundamental frequency as a stress biomarker finds that voice-only systems struggle to separate stress from other aroused states and argues that complementary physiological measures such as HRV are likely necessary for reliable discrimination [8]. This directly motivates the cross-modal design adopted here. Nearest to this work, EMOVOME recognizes emotion from spontaneous voice messages and reports that spontaneous speech trails acted corpora [18].

A further strand of work looks beyond classifying a single state towards tracking change over time, which is precisely what a before-and-after design requires. Speech-derived features have been shown to reflect shifts in psychological state that self-report can miss, which supports measuring the pre-to-post difference rather than only an absolute reading. At the same time, this literature is dominated by English and by acted or laboratory speech, so its direct applicability to spontaneous, multilingual, real-world stress is the open question that the present work examines.

2.3.2 Multimodal and cross-modal stress recognition

Combining signal sources improves reliability over any single modality. The MuSE [24] and WESAD [25] datasets show that physiological and behavioural signals capture complementary dimensions of stress, with WESAD pairing wrist and chest sensors with self-report labels. Together with the fundamental-frequency meta-analysis [8] and VR HRV biofeedback evidence [5], these works support comparing a voice-derived stress reading against an HRV reference and treating divergence between them as informative rather than as error.

A consistent message from the multimodal literature is that modalities fail in different places, so their combination is more robust than anyone alone. Physiological signals respond to autonomic arousal but are slow to reflect the cognitive and affective quality of stress, and they can be confounded by physical activity. Voice responds quickly to affect but as shown later,

conflates stress with other aroused states and is unreliable on the arousal axis for quiet stress. A design that lets each modality contribute to the axis it supports, rather than forcing to carry both, is therefore better matched to how the signals behave, which is the principle behind the confidence-gated cross-modal layer of this component.

2.3.3 Multilingual and low-resource SER, and anomaly detection

Multilingual speech emotion recognition has been approached with transformers trained under augmentation and cross-corpus fusion [19], and cross-lingual pretraining helps languages seen in training [20]. Under-resourced languages such as Sinhala still fall out of distribution for encoders trained mostly on high-resource speech [21], [22], [23]. For session-level outcome evaluation, unsupervised anomaly detection through autoencoder reconstruction error offers a principled way to flag unusual sessions without labelled failure examples [26].

For a language such as Sinhala, which is largely absent from the pretraining corpora of mainstream speech encoders, the model operates in a genuine zero-shot or near-zero-shot regime. Cross-lingual pretraining transfers some structure, but the affective cues that matter stress may not survive for a language the encoder never heard. This thesis treats Sinhala not as a solved multilingual case but as a controlled probe of how a frozen out-of-distribution encoder behaves, and reports the result honestly, including a case adding in-language data did not help.

2.4 Existing systems products and techniques

Three families of existing systems are relevant. Commercial VR meditation platforms such as TRIPP have reached large user bases and features in clinical research [4] but evaluate effectiveness through questionnaires or laboratory equipment rather than automated signal analysis. Wearable physiological monitors capture autonomic stress in real time [25], but rely solely on body-level signals and cannot by themselves confirm a change in subjective stress. Voice-based research systems range from prosody-centred pipelines [17] to pretrained-embedding fusion such as EMOVOME [18], but are typically monolingual, are not embedded in a VR session-validation loop and are not cross-validated against a physiological channel. Table 1 summarized these gaps.

2.4.1 Decision-support rather than autonomous evaluation

A common thread across commercial VR wellness platforms is that they act as decision-support experiences rather than systems that automatically measure whether a session reduced stress, so the effectiveness check still rests on self-report or laboratory equipment.

2.4.2 The single-modality limitation

Wearable monitors capture autonomic arousal but miss the cognitive quality of stress, while voice-only systems struggle to separate stress from other aroused states, which is why this component combines the two under a confidence gate rather than relying on either alone.

2.5 Gap synthesis and conceptual framework

Two kinds of gaps emerge. The research gap is diagnostic: although frozen emotion embeddings are standard, it has not been tested whether ranking them by in-domain acted

agreement predicts real-voice accuracy, nor established which affective axis survives the shift to real spontaneous stress, nor whether any asymmetry is specific to emotion pretraining. The applied gap is systemic: no VR meditation system measures stress from the voice before and after a session, cross-validates it against HRV, and detects unusual sessions within one multilingual component.

These gaps lead to the conceptual framework that organises the thesis. If a frozen emotion encoder preserves the valence of real stress but not its arousal, then stress should be read from valence, and the unreliable arousal magnitude should come from a physiological channel rather than the voice. This yields four design commitments: score stress with a valence-primary rule on an emotion-pretrained encoder; treat the within-speaker change as the primary signal; cross-validate against Component B's HRV under a confidence gate; and flag sessions that deviate from the normal pattern with an autoencoder.

2.5.1 Summary of the identified gaps

In summary, the literature leaves open whether frozen emotion encoders can be selected for real-voice deployment by in-domain agreement, which affective axis they transfer, and how such a model behaves on an out-of-distribution language, and no existing VR system unifies voice stress scoring, HRV cross-validation, and anomaly detection in one multilingual component.

3 RESEARCH METHODOLOGY

3.1 Research design

This study adopts the Design Science Research (DSR) methodology, which centres on constructing and rigorously evaluating a technical artefact that addresses a clearly identified problem. DSR is appropriate here because the deliverable is a working component, yet the central contribution is an empirical finding about that component's core model. The work therefore combines artefact construction with a controlled empirical study, an ablation across model checkpoints and a large-sample replication, as illustrated in Figure 2.

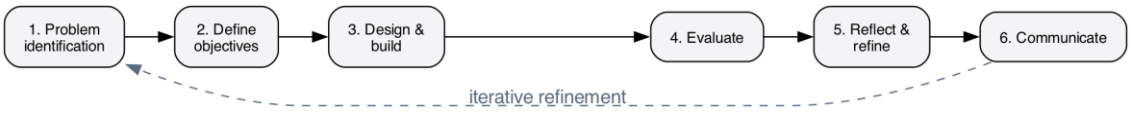


Figure 2: Design Science Research iteration cycle followed in developing and evaluating Component D.

The empirical study is deliberately separated into effect estimation on a small real-voice set and statistical inference on a large independent set, so that an underpowered sample never carries a significance claim.

The Design Science Research cycle is applied iteratively rather than as a single pass. An initial design was built and evaluated, the real-voice evaluation surfaced the arousal-transfer problem, the diagnosis led to a revised scoring rule, and the revised artefact was re-evaluated. Each loop produced both a working improvement to the artefact and a piece of transferable knowledge, which is the dual output that Design Science Research is intended to deliver.

3.2 Data participants or study materials

3.2.1 Real-voice evaluation set

The primary deployment set consists of 24 held-out clips (17 stressed, 7 calm) from consenting speakers, comprising 13 English and 11 Sinhala clips from roughly three Sinhala speakers. All splits are speaker-independent, so no speaker appears in both training and evaluation. The set is intentionally realistic rather than large, and its small size is addressed through leave-one-out thresholding and replication on an independent corpus.

3.2.2 Replication corpus, training corpora, and checkpoints

For large-sample inference, 2,132 held-out utterances from the IEMOCAP corpus are used, each carrying continuous valence and arousal labels. The fusion head is trained on acted English emotion corpora (RAVDESS, CREMA-D, TESS) and on the naturalistic conversational MELD corpus, depending on the checkpoint, and a controlled study additionally mixes 26 locally collected Sinhala clips from seven speakers. Six checkpoints differ only in their training corpus and so span the acted-to-natural range, as summarised in Table 2.

Checkpoint	Training corpus	Role in the study

Fusion-Acted	Acted emotion (RAVDESS, CREMA-D, TESS)	Acted end of the range
Fusion-V2	Acted and real English	Intermediate
Fusion-Combined	Combined acted and natural	Intermediate
Fusion-Augmented	Augmented training	Intermediate
Fusion-IEMOCAP	IEMOCAP (in-domain)	Upper-bound reference only
Fusion-MELD (shipped)	Naturalistic conversational (MELD)	Deployed model

Table 3: The six fusion checkpoints and their role in the study.

Table 4 summarizes the corpora used across training, replication, and evaluation. The acted corpora provide clean, balanced emotional speech that is easy to fit but unrepresentative of spontaneous delivery; MELD provides noisier but far more realistic conversational speech; IEMOCAP provides continuous valence and arousal labels that allow predictions to be correlated against ground truth at scale; and the held-out real-voice set provides the decisive out-of-domain test in both English and Sinhala.

Corpus	**Type**	**Language**	**Role**
RAVDESS	Acted emotional speech	English	Acted training
CREMA-D	Acted emotional speech	English	Acted training
TESS	Acted emotional speech	English	Acted training
MELD	Naturalistic conversational speech	English	Natural training (shipped model)
IEMOCAP	Acted and improvised dyadic, continuous valence/arousal labels	English	Large-sample replication
Real voice set	Spontaneous and elicited stress	English and Sinhala	Held-out deployment evaluation

Table 4: Corpora used across training, replication, and evaluation.

3.3 Experimental or study procedure

3.3.1 Inversion test (RQ1)

Each non-reference checkpoint is scored on the real-voice set under the conventional arousal-based rule, and its real-voice accuracy is related to its in-domain acted arousal agreement. A monotone negative relationship would indicate that the conventional selection criterion is inverted on real stress.

All recordings pass through one preprocessing pipeline before any model sees them. Audio is resampled to 16 kHz mono, high-pass filtered at 70 Hz to remove low-frequency rumble, trimmed of leading and trailing silence, and loudness-normalised to a fixed root-mean-square level. Applying the identical pipeline at training and inference removes the recording-level loudness confound that would otherwise let the model key on microphone gain rather than on affect, which is a common and subtle source of over-optimistic results in speech-emotion work.

3.3.2 Axis-split analysis and replication (RQ2)

For every checkpoint, the proportion of real stressed clips on which valence is placed on the correct side is compared against the proportion on which arousal is placed on the correct side. The split is then replicated on the 2,132-utterance IEMOCAP set by correlating predicted valence and arousal with ground truth and testing the per-utterance difference with a bootstrap confidence interval.

3.3.3 Encoder-specificity probe (RQ3)

A matched-size ridge probe is trained on IEMOCAP embeddings from emotion2vec and from WavLM at equal representation width, isolating whether any valence advantage is an effect of emotion pretraining rather than model capacity.

3.3.4 Scoring-rule comparison and deployment (RQ4, RQ5)

The valence-primary rule is compared against the legacy arousal-based rule by the separation it produces every checkpoint, and the shipped checkpoint deployment accuracy is measured under leave-one-out thresholding, reported overall and per language.

The training configuration is shared across all six checkpoints so that only the training corpus varies, isolating its effect. Table 5 lists the configuration. The frozen encoder produces a 1024-dimensional embedding; the prosodic branch produces 21 features, both are projected to a common width of 128 and combined by a learned gate, and a small head regresses valence and arousal with a hyperbolic-tangent output. Training minimises a concordance loss with a small mean-squared-error term, and the same preprocessing is applied at training and inference, so there is no distribution shift introduced by the pipeline itself.

Setting	Value
Encoder	emotion2vec-plus-large (frozen, about 300M parameters)
Embedding dimension	1024
Prosodic features	21 interpretable features
Projection width	128 (embedding and prosody)
Head	128 to 64 to 2, ReLU, dropout 0.3, tanh output
Trainable parameters	about 1 to 2 million (head, gate, projections)
Loss	1 minus the average of valence and arousal CCC, plus 0.2 times MSE
Optimizer	Adam
Preprocessing	16 kHz mono, 70 Hz high-pass, silence trimming, loudness RMS 0.05
Stressed threshold	score greater than 2.0 (leave-one-out validated)
Confidence gate	defer to HRV below confidence 0.4

Table 5: Shared training and inference configuration across all checkpoints.

3.4 Measures and evaluation metrics

The study uses metrics appropriate to each claim rather than a single headline number, summarised in Table 3.

Metric	Purpose
Leave-one-out accuracy	Deployment accuracy with the threshold never tuned on the clip it scores
Concordance correlation coefficient (CCC)	Valence and arousal regression quality (penalises scale and location error)
Pearson correlation	Predicted versus true axis values on IEMOCAP
Separation (d-prime)	Threshold-free class-mean gap divided by pooled standard deviation
Cohen's d [28]	Effect size for the paired scoring-rule comparison
Spearman rank correlation	Monotone relationship underlying the inversion
Bootstrap 95% intervals [29]	Uncertainty on per-language accuracy and on the valence minus arousal difference

Table 6: Evaluation of metrics and their purpose.

A minimum stress-classification accuracy of 75 percent across English and Sinhala on held-out real data is adopted as the engineering acceptance target, consistent with reported cross-lingual SER baselines.

3.4.1 Deployment accuracy

Deployment accuracy is reported under leave-one-out thresholding, so the decision threshold is never fitted on the clip it scores, and it is reported overall and separately for English and Sinhala with bootstrap confidence intervals.

3.4.2 Agreement, correlation, and separation

Regression quality uses the concordance correlation coefficient; predicted axes are correlated with ground truth on IEMOCAP, and the threshold-free separation between stressed and calm speech is measured with d-prime.

3.4.3 Effect size and uncertainty

The paired comparison between scoring rules is summarized with Cohen's d, and bootstrap resampling provides confidence intervals that must exclude zero for the valence over arousal split to be considered significant.

3.5 Data analysis

Because the real-voice set is small, effect estimation and statistical inference are kept apart. On the 24 clips and six checkpoints, the analysis reports effect sizes, rank correlations, and bootstrap intervals, and does not treat underpowered p-values as evidence. Statistical inference is instead carried by the IEMOCAP replication, where the valence over arousal difference is tested with bootstrap confidence intervals that must exclude zero. Deployment accuracy uses leave-one-out thresholding, and per-language estimates are reported with their intervals, so the wide Sinhala interval is visible rather than hidden.

The separation of estimation from inference is a deliberate guard against over-claiming on a small sample. On the 24-clip set, point estimates and effect sizes describe what was observed, but no significance is claimed from them, because a handful of clips cannot support such a

claim. The significance rests entirely on the independent-corpus replication, where the sample is large enough for the bootstrap interval on the valence-minus-arousal difference to be meaningful, and where that interval excludes zero for every out-of-domain checkpoint.

3.6 Ethics privacy security and responsible AI

Voice is personal data, and the component is designed around data minimization. All participants in the real-voice collection gave informed consent covering how recordings would be used, stored, and deleted. Recordings are anonymized by replacing identifiers with numerical codes before processing, and raw audio is not retained for longer than needed to produce a score. The deployed system transmits audio over encrypted connections and keeps third-party service credentials server-side rather than on the device.

The component reports a well-being stress signal and never a medical diagnosis; all user-facing language is framed accordingly, and uncertain readings are surfaced as uncertain. In keeping with responsible-AI practice and module policy, the use of generative-AI and automated tools in preparing this thesis is disclosed in Appendix B, and the author remains responsible for the accuracy and verification of all claims, code, and analyses.

3.6.1 Consent and data minimisation

Participants gave informed consent; recordings were anonymized with numerical codes, and raw audio was retained only as long as needed to produce a score.

3.6.2 Responsible use and fairness

The component reports a well-being signal rather than a medical diagnosis, surfaces uncertain readings as uncertain, and documents its out-of-distribution limits for Sinhala, so its dependable range is not overstated.

3.7 Reproducibility and research artefacts

Reproducibility is built into the design. Training and inference share a single preprocessing pipeline, which removes the recording-level confound that would otherwise differ between train and test. The encoder is frozen, so only the small fusion head varies between checkpoints, and the controlled Sinhala study was retrained under a single fixed random seed so that the label scheme was the only difference between variants. The research artefacts comprise the six fusion checkpoints, the shipped model and anomaly detector, the preprocessing and scoring code, and the evaluation scripts for the real-voice and IEMOCAP studies, described alongside the implementation in Chapter 5.

3.7.1 Controlling randomness

The controlled Sinhala study was retrained under a single fixed random seed so that the label scheme was the only difference between variants, isolating its effect from random initialization.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The component's stakeholders are the end user who completes the before and after voice check-in, the wider project group who's physiological, audio, and VR-adaptation components exchange signals with it, and the supervisor and assessors who evaluate the scientific contribution.

4.1.1 Functional requirements

FR1. Assess the ambient acoustic environment before capturing and gate the session until it is suitable. FR2. Capture a short voice recording before and after the VR session. FR3. Preprocess each recording through a single shared pipeline. FR4. Produce an acoustic stress score and confidence value using a valence-primary rule on a frozen emotion encoder. FR5. Compute the before-to-after-stress change and classify the outcome. FR6. Cross-validate the voice reading against Component B's HRV stress and defer to HRV when voice confidence is low. FR7. Detect sessions that deviate from the normal stress-reduction pattern. FR8. Present a session report. FR9. Operate across English and Sinhala with a language-agnostic scoring path.

Table 6 lists the functional requirements as discrete, testable statements.

ID	Functional requirement
FR1	Assess the ambient environment and gate the session until it is suitable.
FR2	Capture a short voice recording before and after the VR session.
FR3	Preprocess each recording through a single shared pipeline.
FR4	Produce stress scores and confidence value using valence-primary scoring.
FR5	Compute the before-to-after change and classify the outcome.
FR6	Cross-validate the voice reading against HRV and defer to HRV when uncertain.
FR7	Detect sessions that deviate from the normal stress-reduction pattern.
FR8	Present a session report combining all layer outputs.
FR9	Operate across English and Sinhala with a language-agnostic scoring path.

Table 7: Functional requirements of Component D.

4.1.2 Non-functional requirements

The system warms its models on start so first-clip latency is low, degrades gracefully when any external dependency is unavailable, keeps credentials server-side and minimises audio retention, presents a guided flow in plain non-clinical language, and uses configuration-driven thresholds with a frozen-encoder design so the model can be improved without disturbing the surrounding system.

Table 7 lists the non-functional requirements and how the design satisfies each.

Quality	Requirement and how it is met
Performance	Models are warmed on start, so first-clip latency is low.
Reliability	Every external dependency degrades gracefully; the flow never blocks.
Security and privacy	Credentials stay on server-side; audio is encrypted and minimised.
Usability	Guided, non-clinical flow with no manual record control.
Maintainability	Configuration-driven thresholds and a swappable trainable head.

Table 8: Non-functional requirements and how the design satisfies them.

The principal stakeholders are summarized in Table 8.

Stakeholder	Interest
End user / participant	Completes the check-in and receives clear session feedback.
Project group (A, B, C)	Exchange physiological, audio, and VR-adaptation signals with D.
Supervisor and assessors	Evaluate scientific contribution and the artefact.
Researchers and practitioners	Objective, automated session-level stress data.

Table 9: Principal stakeholders of the component.

4.2 Component architecture within the complete solution

Component D is realized as a client-server system. An Android mobile application captures audio and presents results, and a backend service hosts the models and the five-layer pipeline. Within the complete solution, it reads Component B's HRV stress for cross-modal validation and brackets the VR session with the before and after check-ins. Figure 3 shows the internal architecture, and Table 4 lists the principal API endpoints.

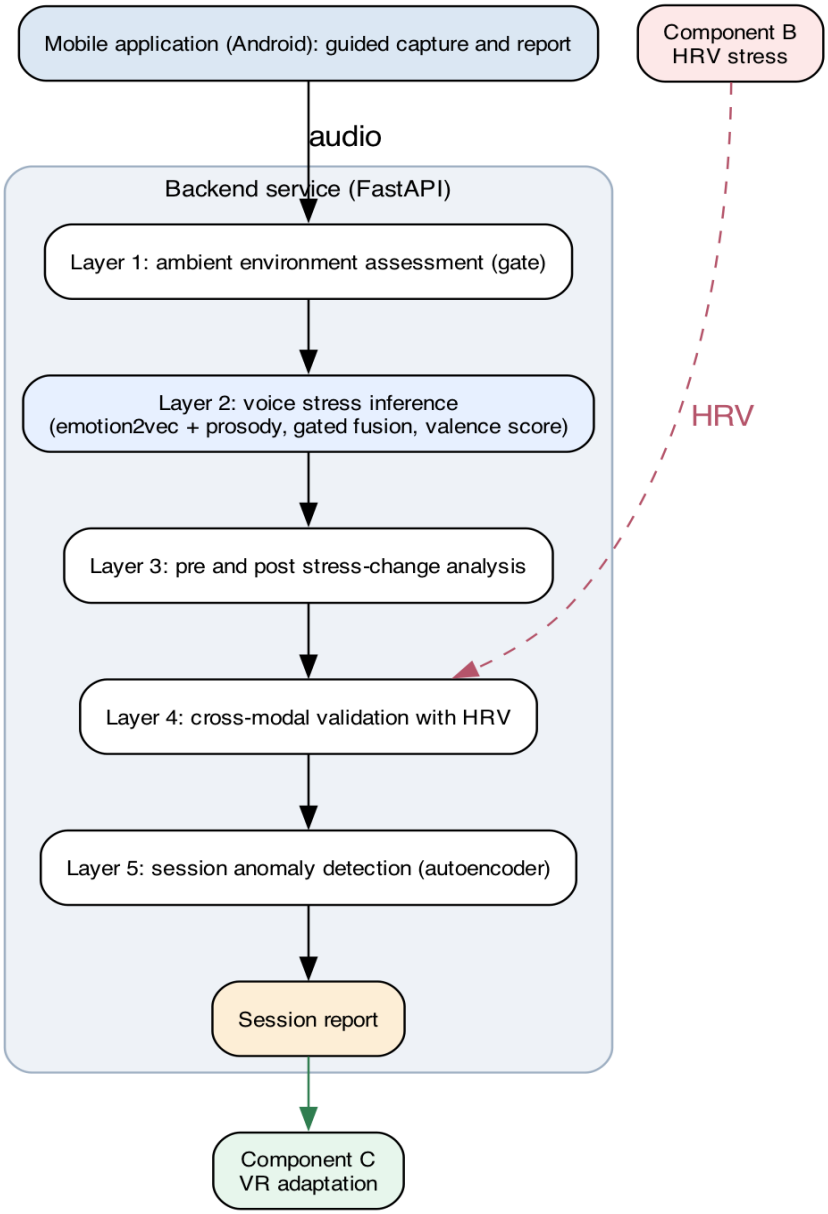


Figure 3: Internal five-layer architecture of Component D and its links to Components B and C.

Endpoint	**Purpose**
GET /health	Readiness of all five layers
POST /warmup	Pre-load models so the first analysis is fast
POST /ambient-check	Layer 1 environment verdict, score, and metrics
POST /companion/voice-turn	One guided turn: transcribe, reply, and on the final clip score and store
GET /companion/tts	Companion voice audio, with on-device fallback
POST /session-update	Entry point for Component B to push an HRV reading
POST /full-session	Run Layers 3 to 5 and return the combined report

Table 10: Principal backend API endpoints.

Data flows through the component in a fixed order. The user speaks, the mobile client streams the audio to the backend, Layer 1 decides whether the environment is usable, and only then does scoring proceed. Before reading is stored, the user enters the VR session, and on return the after reading is captured and scored the same way. The full-session step then reloads both readings, computes the change, reconciles the voice reading with the physiological one, checks the session against the normal pattern, and returns a single report. Because each reading has persisted as soon as it is produced, a crash or restart between the before and after phases does not lose work.

The split between a thin mobile client and a model-hosting backend is intentional. Running a 300-million-parameter encoder on a phone would be slow and battery-hungry, so the device handles capture, the guided conversation, and presentation, while the backend handles the heavy inference behind a small, well-defined interface. This also keeps third-party credentials and model weights off the device, which is important for both security and maintainability.

4.3 Design alternatives and decisions

The design reflects a sequence of evidence-based decisions, the most consequential following directly from the empirical study. Table 5 records the principal alternatives considered.

Decision	Alternatives	Choice and rationale
Encoder	XLSR-53 or wav2vec2 versus emotion2vec	emotion2vec: outperformed wav2vec2 in the ablation (84.7% versus 60.9%) and carries valence far better at matched size
Scoring rule	Arousal-based versus valence-primary	Valence-primary: arousal does not transfer to real stress, and valence-primary widens separation on every checkpoint
Primary signal	Absolute score versus within-speaker changes	Within-speaker change cancels speaker-identity bias
Arousal magnitude	Infer from voice versus defer to HRV	Defer to Component B: the voice does not supply arousal dependably
Capture	Manual record button versus guided automatic capture	Guided capture: reduces friction and yields more natural speech

Table 11: Principal design alternatives and decisions.

4.4 Detailed design

4.4.1 Layer 1: ambient environment assessment

A short sample is analyzed for its noise floor and returns a three-state verdict: good at or below -45 dBFS, usable between -45 and -30 dBFS, and too noisy above -30 dBFS, together with voice-activity detection for other speakers and a check for sudden peaks. The layer acts as a gate before any scoring.

4.4.2 Layer 2: voice stress inference

Audio passes through a frozen emotion2vec encoder of about 300 million parameters, used only as a feature extractor, producing a 1024-dimensional embedding, and in parallel through a prosodic front-end that computes 21 interpretable features (pitch statistics, jitter, shimmer, harmonics-to-noise ratio, RMS energy, zero-crossing rate, spectral centroid, roll-off and flatness, speech rate, pause ratio, voiced fraction, and duration). A gated fusion head, the only trainable part, mixes the two streams and regresses valence and arousal. Figure 4 shows the model, and the fusion is defined as follows, where e and p are the projected embedding and prosody streams and g is a learned gate in the open interval zero to one:

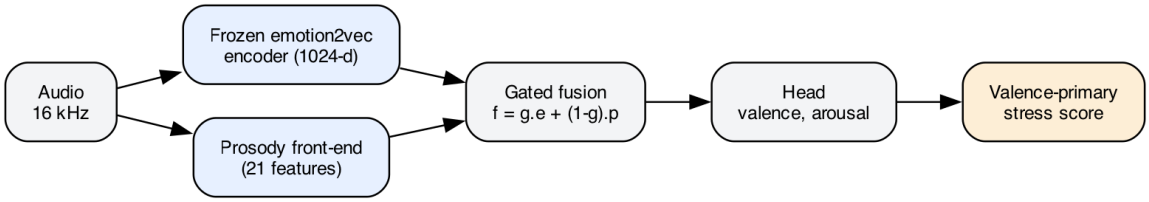


Figure 4: Layer 2 model: frozen emotion2vec and a trainable prosody branch combined by a gated fusion head.

$$f = g.\ e + (1 - g).\ p$$

$$(v, a) = \tanh(\text{head}(f))$$

Only the head, the gate, and the two projections are trained, on the order of one to two million parameters, while the encoder stays frozen. Training minimizes one minus the average of the valence and arousal concordance coefficients plus a small mean-squared-error term.

The prosodic branch computes 21 interpretable features grouped as in Table 9, chosen in the spirit of a minimalistic acoustic parameter set so that each feature is both stress-relevant and explainable. Undefined values returned by the pitch analyzer are replaced with safe defaults, so the model never receives invalid input.

Group	Features
Pitch	Fundamental-frequency mean, standard deviation, minimum, and maximum
Voice quality	Jitter, shimmer, harmonics-to-noise ratio
Energy	Root-mean-square energy, zero-crossing rate
Spectral shape	Spectral centroid, roll-off, flatness
Timing	Speech rate, pause ratio, voiced fraction, duration

Table 12: The prosodic feature groups computed by the Layer 2 front-end (21 features in total).

4.4.3 Valence-primary scoring

The deployed rule reads stress from valence rather than arousal, as shown in Figure 5. The stress score is ten times the negative part of predicted valence, and confidence is the absolute value of valence:

$$\text{Stress} = 10.\ \max(0, -v),\ \text{confidence} = |v|$$

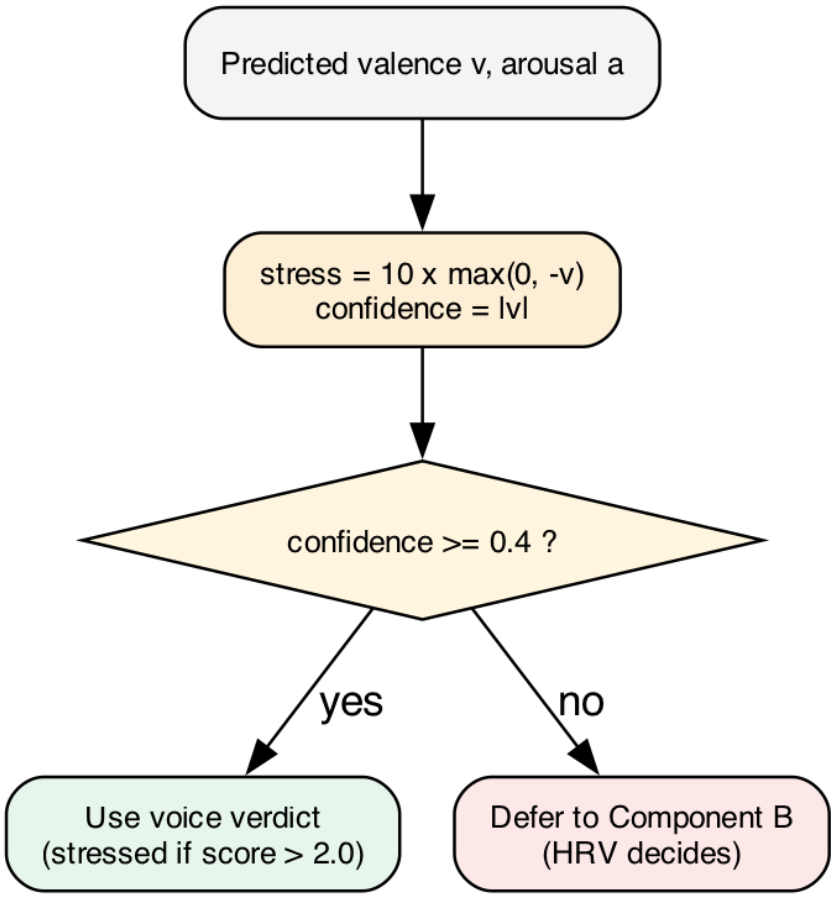


Figure 5: Valence-primary scoring and the confidence gate that defers to HRV when the voice is uncertain.

A clip is flagged stressed above a score of 2.0; a threshold set by leave-one-out validation. Arousal magnitude is deliberately not used as a stress magnitude and is handed to Component B.

Reading stress from valence rather than arousal is not an arbitrary choice but the direct consequence of the empirical finding in Chapter 6. Because the encoder preserves valence for real stress but loses arousal, a rule that derived stress magnitude from arousal would be built on the one axis the encoder cannot supply, whereas a rule built on valence rests on the axis it can. Defining confidence as the magnitude of valence then makes the system least confident exactly on the ambiguous clips where valence is near zero, which is the right behaviour for a signal that should defer when it is unsure.

4.4.4 Layers 3 and 4: change analysis and cross-modal validation

Layer 3 computes the absolute and relative change between the before and after scores and classifies the outcome as reduced, unchanged, or increased, with a reliability of judgement. Layer 4 reconciles the voice reading with Component B's HRV stress, as shown in Figure 6. It maps B's ordinal levels into the component's scale, computes agreement, and below a confidence floor of 0.4 defers to HRV rather than asserting a potentially wrong verdict.

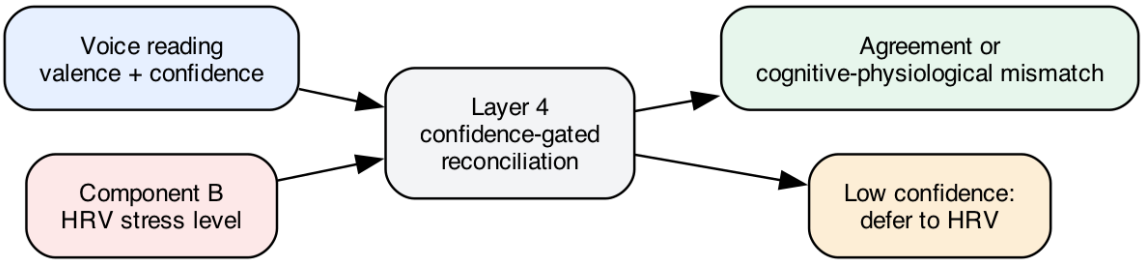


Figure 6: Cross-modal validation: voice valence and HRV arousal are reconciled under a confidence gate.

Component B reports on an ordinal stress level, either as a point estimate or as a band spanning two adjacent levels, each with a confidence value. The cross-modal client maps these into the component's four-level vocabulary as in Table 10, taking the higher level for a band so stress is never under-called, and capping a band's confidence so an uncertain physiological reading defers rather than overrides. A malformed or unavailable reading is treated as no body signal, so the system falls back to voice-only rather than fabricating a value.

Component B reading	**Mapping in Component D**
Point level 0 to 3	Mapped directly to no, mild, moderate, or high
Band (two adjacent levels)	Higher of the two levels; confidence capped
Low confidence (below 0.4)	Layer 4 defers to the physiological estimate
Malformed or unavailable	Treated as no body signal; voice-only fallback

Table 13: Mapping of Component B HRV readings into Component D cross-modal validation.

4.4.5 Layer 5: session anomaly detection

A session-level feature vector is passed through an autoencoder, and a session whose reconstruction error exceeds the mean plus three standard deviations of the normal distribution is flagged, switching to a per-user threshold once enough of that user's sessions are available. The direction of any anomaly is labelled, so an unusually good session is not mis-reported as an alarm.

4.4.6 Preprocessing

Training and inference share one pipeline: resampling to 16 kHz mono, a 70 Hz high-pass filter, silence trimming, and loudness normalization to a fixed RMS of 0.05. Because normalization amplifies a very quiet clip and its noise floor, a faint-recording guard reduces confidence in proportion to how far the input falls below an RMS of 0.02, and a quality gate drops clips that are too quiet, clipped, or short on detected voice before scoring.

Two guards protect the score against degenerating input. The faint-recording guard recognizes that loudness normalization amplifies a very quiet clip together with its noise floor, which can turn a near-silent recording into a confident but meaningless reading; it therefore reduces confidence in proportion to how far the raw input falls below a minimum level. The quality gate rejects clips that are clipped, too short on detected voice, or otherwise unusable before they reach the model, so that the user is asked to try again rather than being given a spurious result.

4.4.7 Companion-guided mobile experience

Beyond the five analytical layers, the component is delivered as a companion-guided mobile experience, and the user-facing design is itself a contribution to data quality. The check-in is presented as a short conversation with a named companion across a small number of stages: an introduction and consent, a room check, the before check-in, the VR hand-off, the after check-in, and the report. Recording happens automatically in the background, the companion asks a few short, easy questions, so the user keeps speaking naturally, and every answer in a phase is merged into one clip that is scored once. The companion always speaks, falling back to on-device speech if the server-side voice is unavailable, so the conversation never stalls.

4.5 Standards and professional practice

The component follows established conventions. Client-server communication uses REST over HTTP with JSON payloads and explicit status codes, so an unavailable model or service returns a clean, documented error rather than a crash. The dimensional-emotion evaluation follows standard practice, and the thesis uses IEEE referencing as recommended for the module. Professional practice is reflected in privacy by design, accessibility and multilingual support, responsible-AI conduct, and maintainability through configuration-driven thresholds and a swappable trainable head.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The backend is implemented in Python with FastAPI, served by Uvicorn, and uses PyTorch for all deep-learning components. The frozen emotion2vec encoder is obtained through ModelScope, speech-to-text uses faster-whisper, the guided companion's reply text is produced by a local open model served through Ollama, and the companion's voice is synthesised through ElevenLabs, proxied server-side. The prosodic front-end uses Praat (via parselmouth) and librosa. Table 6 lists the stack.

Area	Technology
Backend and API	Python, FastAPI (Uvicorn)
Deep learning	PyTorch
Voice encoder	emotion2vec (frozen, via ModelScope)
Prosodic features	Praat (parselmouth) and librosa (21 features)
Speech-to-text	faster-whisper
Companion replies	Ollama (local open model)
Companion voice	ElevenLabs (server-side proxy, on-device fallback)
Mobile app	React Native (Android, Kotlin native audio modules)
Trained artefacts	fusion_meld_baseline (Layer 2), anomaly detector (Layer 5)

Table 14: Implementation technology stack.

5.1.1 Backend stack

The backend is a Python FastAPI service hosting the frozen emotion2vec encoder, the prosodic front-end, speech-to-text, and the companion reply and voice services behind a small HTTP and JSON contract.

5.1.2 Mobile stack

The mobile client is a React Native application for Android, with audio capture, playback, concatenation, and file selection implemented as native Kotlin modules for reliable on-device recording.

5.2 Component implementation

Each layer is implemented behind the HTTP contract of Section 4.2. Layer 2 loads the frozen encoder once and keeps it resident, computes the embedding and the prosodic features through the shared preprocessing pipeline, runs the gated fusion head, and applies valence-primary scoring with the faint-recording guard. Scored readings have persisted to disk, so a mid-session restart does not lose the before result, and the full-session report can reload both phases.

The guided check-in is served by a single conversational-turn endpoint that, for each turn, transcribes the clip, produces a companion reply, and only on the final clip of a phase score and stores it. This lets the application hold a short multi-turn conversation and score once on the concatenated audio. A speech-to-text hallucination guard prevents a spurious transcript

from influencing the stress score, and the companion voice falls back to on-device speech whenever server-side synthesis is unavailable, so the flow never blocks.

Audio arriving at the backend is decoded in two stages, a fast path for uncompressed audio and a fallback for compressed container formats, so the same scoring code accepts whatever the device records. The frozen encoder is loaded once and kept resident in memory, which keeps per-request latency low, and a warmup endpoint triggers this load before the first real clip so that the first user does not pay the loading cost. Each scored reading is written to disk immediately, keyed by session and phase, so the full-session report can reload both phases even after a restart.

The guided conversation is handled by a single endpoint that, for each turn, transcribes the clip, generates a companion reply, and, only on the final clip of a phase, scores and stores the merged audio. A speech-to-text hallucination guard ensures that a nonsense transcript, which can occur for very short or noisy input, never influences the acoustic score, and the reply model being unavailable never fails a turn, because the turn's real value, the transcript and the Layer 2 reading, is computed independently of the reply.

5.3 Integration with the other components

Component B runs its own continuous HRV stress stream and exposes its latest reading. Because Component D is episodic, it reads B's stress at the two moments the user speaks and feeds the value into Layer 4. Figure 7 shows the integrated session sequence, with B polled at each phase.

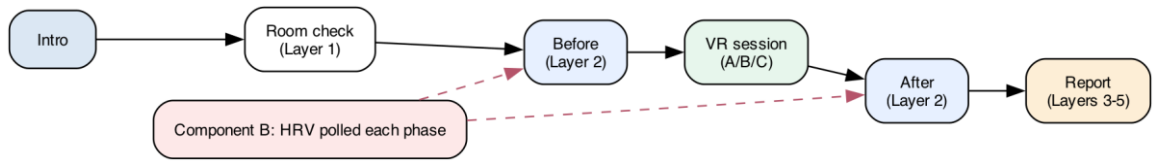


Figure 7: Integrated session sequence across the before and after phases, with Component B polling each phase.

Two routes converge on the same stored representation: Component D can poll B's latest-reading endpoint at scoring time, and Component B can push a reading to Component D's session-update endpoint. A dedicated client maps B's ordinal levels and confidence into the component's vocabulary, treating an uncertain band reading conservatively, so a low-confidence physiological reading defers rather than asserts.

Stated honestly, the cross-modal client and the Layer 4 reconciliation are implemented and unit-tested, and the pipeline has been validated end-to-end using stored and mock HRV readings that exercise the same code path a live reading follows. A full end-to-end run against a live Component B with a streaming wearable remains to be exercised and is identified as integration work in Chapter 7. Component D also brackets the VR session, handing off to the VR experience after the before check-in and resuming for the after check-in, and the full-session report provides the session outcome that Component C consumes.

5.4 Verification and software testing

The backend is covered by an automated test suite spanning the layers and the integration boundary, summarised in Table 7. Beyond unit tests, the system was verified end-to-end through the live API, scoring a before and after pair then running comparison, cross-validation, and the full-session report, and the health endpoint confirms that all five layers are loaded. The mobile application was kept type-clean and exercised on real Android devices across the full guided flow, including the companion voice path and the multi-turn capture-and-score path.

Test area	Representative tests
Layer 1 quality	Ambient verdict boundaries and voice-activity checks
Layer 2 model and prosody	Scoring behaviour, faint-recording penalty, stress mapping
Layer 3 comparison	Before and after change and reliability
Layer 4 cross-modal	Confidence gate and conservative band handling
Layer 5 anomaly	Reconstruction-error flagging and per-user threshold
Component B client	Success, not-ready, unreachable, and malformed responses
API and companion	Endpoint behaviour and turn logic
Preprocessing and storage	Shared pipeline, session store, persistence

Table 15: Automated test coverage by area.

5.4.1 Unit testing

Unit tests cover the deterministic components, including prosodic feature extraction, the preprocessing pipeline, the stress mapping, and the Component B schema mapping, where exact outputs can be asserted.

5.4.2 End-to-end and on-device testing

The full chain was verified live through the API, and the mobile application was exercised on physical Android devices across the entire guided flow, including the companion voice path and the multi-turn capture-and-score path.

Table 11 lists the automated test modules and what each verifies, spanning the five layers, the Component B client, the API and companion logic, and the shared preprocessing and storage.

Test module	Verifies
test_layer1_quality	Ambient verdict boundaries and voice-activity detection
test_layer2_fusion / test_layer2_prosody	Fusion scoring and prosodic feature extraction
test_stress_mapping	Valence-primary score and confidence mapping
test_layer3_compare	Before-and-after change and reliability
test_layer4_confidence / test_layer4_crossmodal	Confidence gate and band handling
test_layer5_anomaly / test_personal_baseline	Reconstruction-error flagging and per-user threshold
test_component_b_client	Success, not-ready, unreachable, and malformed responses
test_api / test_companion	Endpoint behaviour and conversational-turn logic

test_preprocessing / test_store / test_session_log	Shared pipeline, session store, and persistence

Table 16: Automated test modules and what each verifies.

5.5 Deployment and operation

The backend is deployed as a local FastAPI service started with Uvicorn. A warmup endpoint pre-loads the models, so the first analysis is fast, and the large model weights download to a local cache on first run. The companion reply model is served locally through Ollama, and the voice-synthesis credential is supplied only through the server environment, never shipped in the application. A missing credential or an unreachable dependency degrades gracefully to the on-device voice and does not interrupt scoring. For a cable-free demonstration, the application communicates with the backend over a wireless developer bridge, and deployment to a managed cloud environment is a straightforward extension noted as future work.

In operation, the backend is a single service that warms its models on start and then serves scoring, companion, and full-session requests. The large model weights are downloaded once to a local cache, so the first launch on a new machine needs network access, but subsequent launches do not. The companion reply model runs locally, and the voice-synthesis credential is supplied only through the server environment and never shipped in the application, so a leaked build cannot expose it.

For demonstrations without a cable, the device communicates with the backend over a wireless developer bridge, with the service and bundler ports tunnelled to the phone. The most common operational issue is the loss of this bridge when the screen locks or the network changes, which is resolved by re-establishing the tunnels; the service itself is unaffected because each external dependency has a defined fallback. Migration to a managed cloud deployment is a configuration change rather than a redesign, since the component already communicates over a small HTTP interface.

6 RESULTS AND VALIDATION

6.1 Research results

The research results answer the five research questions: the selection-criterion inversion (RQ1), the valence and arousal split and its large-sample replication (RQ2), the encoder-specificity of that split (RQ3), and the scoring-rule improvement (RQ4). Deployment behaviour across languages (RQ5) is reported with the product results in Section 6.2.

6.1.1 The inversion (RQ1)

Relating each checkpoint's in-domain acted arousal agreement to its real-voice accuracy under the conventional rule yields a monotone negative relationship, with a Spearman rank correlation of -0.60, as shown in Figure 8 and Table 8. The weakest in-domain checkpoint deploys best, and the strongest in-domain checkpoint deploys worst. With five distinct in-domain points, this correlation is directional rather than significant, but the ordering is unambiguous and is explained by the mechanism below.

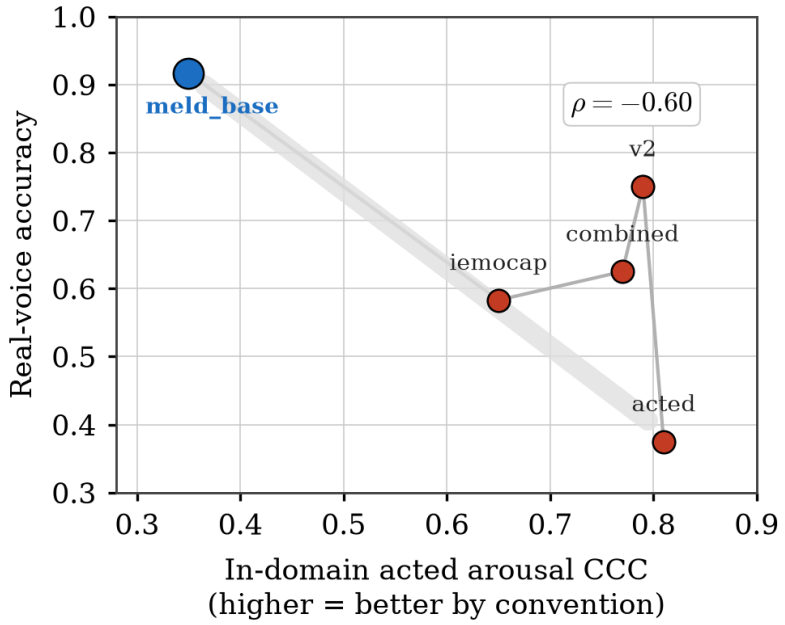


Figure 8: The inversion: real-voice accuracy falls as in-domain acted arousal agreement rises.

Checkpoint	In-domain acted CCC	Real-voice accuracy
Fusion-Acted	0.81	0.375
Fusion-V2	0.79	0.750
Fusion-Combined	0.77	0.625
Fusion-IEMOCAP	0.65	0.583
Fusion-MELD (shipped)	0.35	0.917

Table 17: The inversion: higher in-domain agreement coincides with lower real-voice accuracy (Spearman rho of -0.60).

The practical consequence of this inversion is that the conventional model-selection step, choosing the checkpoint with the best in-domain agreement, would in this setting select the

worst deployable model. The checkpoint trained on naturalistic conversational speech, which looks weakest on acted benchmarks, is the one that generalises to real stress. This is why the shipped model is Fusion-MELD rather than the acted-trained checkpoint that the convention would prefer.

6.1.2 The valence and arousal split (RQ2)

Separating the two axes exposes the cause, shown in Figure 9. On real stressed clips, valence is placed on the correct (negative) side 82 to 100 percent of the time, reaching 100 percent for the shipped checkpoint, while arousal is placed on the correct (high) side only 12 to 53 percent of the time. Real stress reliably reads as negative in valence but fails to read as high in arousal, because the quiet freeze of genuine stress falls in the low-arousal region that acted training never emphasises.

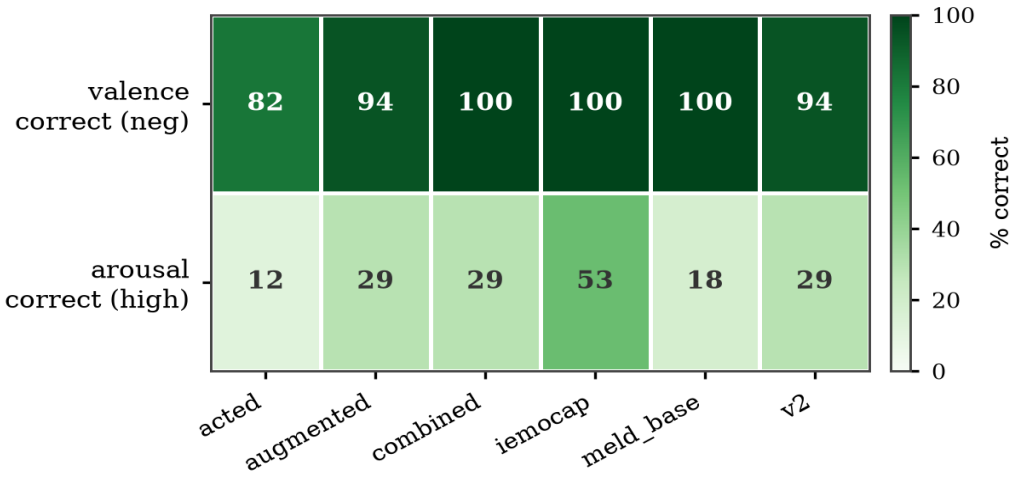


Figure 9: Valence is recovered (82 to 100 percent) while arousal is not (12 to 53 percent) on real stressed clips.

The asymmetry is visible at the level of individual clips. A quiet, flattened stressed voice is read as clearly negative in valence, which is correct, but as low in arousal, which the conventional reading would suggest calm. It is precisely this combination, negative valence with low arousal, that the valence-primary rule interprets correctly as shutdown-type stress and that the arousal-based rule misses. The two axes therefore do not fail together; one survives the shift to real stress and the other does not.

6.1.3 Replication on IEMOCAP (RQ2 at scale)

Because 24 clips cannot support a significance claim, the split is replicated on 2,132 IEMOCAP utterances by correlating predicted valence and arousal with ground truth, in Table 9. For every checkpoint trained outside IEMOCAP, valence correlates well above arousal, and the bootstrap 95 percent interval on the per-utterance difference excludes zero in every case, so the split is significant at scale.

Checkpoint	Valence r	Arousal r	Difference (95% CI)
Fusion-MELD (shipped)	0.688	0.485	+0.202 [0.159, 0.244]

Fusion-Acted	0.662	0.326	+0.336 [0.293, 0.381]
Fusion-V2	0.674	0.404	+0.270 [0.225, 0.314]
Fusion-Combined	0.726	0.621	+0.104 [0.072, 0.140]
Fusion-IEMOCAP (reference)	0.742	0.659	in-domain upper bound

Table 18: IEMOCAP replication (2,132 utterances): valence exceeds arousal and the interval on the difference excludes zero.

6.1.4 Encoder specificity (RQ3)

A matched-size ridge probe on IEMOCAP embeddings isolates whether the valence advantage comes from emotion pretraining or from model size, in Figure 10 and Table 10. At equal width, emotion2vec carries valence far better than WavLM (CCC 0.660 versus 0.387), and the direction of the asymmetry is encoder-specific: emotion2vec favours valence whereas WavLM favours arousal. The valence over arousal split is therefore a specific property of emotion pretraining, which is why valence-primary scoring is right for this encoder and would be wrong for WavLM.

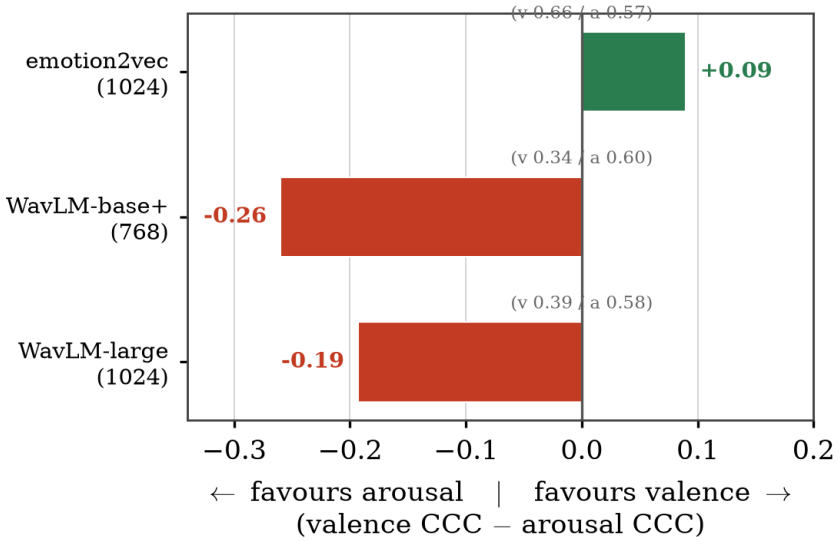


Figure 10: Matched-size probe: emotion2vec favours valence, while WavLM favours arousal.

Encoder	Dimension	Valence CCC	Arousal CCC
emotion2vec-plus-large	1024	0.660	0.570
WavLM-base+	768	0.344	0.604
WavLM-large	1024	0.387	0.580

Table 19: Matched-size ridge probe on IEMOCAP.

6.1.5 The fix: valence-primary scoring (RQ4)

Acting on the diagnosis, the valence-primary rule is compared against the conventional arousal-based rule by the separation it produces on every checkpoint, in Figure 11. Valence-primary scoring improves separation on all six checkpoints, with a mean gain of about 0.50 and a large effect (Cohen's d of 1.71). For the shipped checkpoint, the separation rises from 1.86 to 3.62.

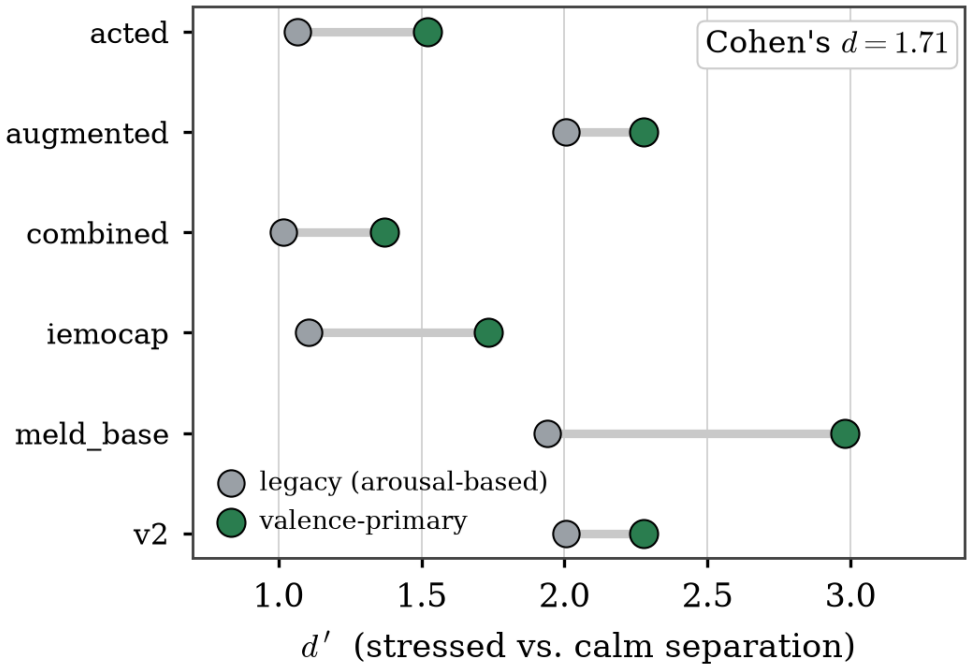


Figure 11: Valence-primary scoring improves the stressed and calm separation on every checkpoint (Cohen's d of 1.71).

The improvement is consistent in direction on every checkpoint, which is the appropriate claim for six paired observations, and it is large in size. Crucially, the gain does not come from tuning to the real-voice set, because the rule follows from the diagnosis rather than from a search over the evaluation data, and the deployment threshold is set by leave-one-out validation so that no clip informs its own decision. The fix is therefore a principled correction rather than an overfit to the small sample.

6.2 Product and technical results

6.2.1 Deployment accuracy

Under leave-one-out thresholding, the shipped checkpoint reaches 91.7 percent on the real-voice set, in Table 11. By language, English separates without error in this sample and Sinhala reaches 0.909. The wide Sinhala interval follows directly from the small sample and roughly three speakers, so the point estimate is not over-read. Both languages meet the 75 percent acceptance target.

Split	n	Accuracy	95% CI
All (leave-one-out)	24	0.917	[0.875, 1.000]
English	13	1.000	[1.000, 1.000]
Sinhala	15	0.909	[0.727, 1.000]

Table 20: Deployment accuracy of the shipped model (overall is leave-one-out; per-language values are fixed-threshold with bootstrap intervals).

6.2.2 Controlled Sinhala study

A controlled test of whether more in-language data helps produce a negative result: adding 26 Sinhala training clips lowered held-out Sinhala accuracy from 90.9 to 81.8 percent. Retraining

binary-labelled and graded variants under a single fixed seed landed at the same 81.8 percent, so the label scheme only relocates the error, and the regression is attributable to the Sinhala data passing through a frozen out-of-distribution encoder rather than to labelling.

6.2.3 End-to-end verification

The complete pipeline was verified end-to-end through the live API on a real before and after pair. Layer 3 reported a reliable, strong improvement, Layer 4 validated the voice reading against the stored HRV value with an agreement of 0.842, and Layer 5 labelled a large positive swing as an unusual improvement rather than raising it as an unqualified alarm, confirming that the five layers operate together.

Taken together, the per-language results, the controlled Sinhala study, and the end-to-end verification show a system that is strong and dependable on the language it was pretrained for, honest about its limits on an out-of-distribution language, and coherent across its five layers. The deployment accuracy is not an isolated number but the end point of a chain of evidence that begins with the diagnosis of which affective axis transfers and ends with a working, cross-validated session report.

6.3 User or stakeholder validation

6.3.1 On-device validation

The component was exercised on real Android devices across the full guided check-in, confirming that the companion-led flow captures usable audio automatically, that the room-check gate behaves as intended, and that the voice path degrades gracefully to on-device speech when server-side synthesis is unavailable. The guided, non-clinical presentation and the absence of a manual record control were the design responses to the usability and comfort concerns raised in the requirements.

6.3.2 A confident single speaker misread

A specific limiting observation is reported honestly. In a small single-speaker probe of three clips of Sri-Lankan-accented English, the model read a genuinely calm, post-meditation clip as strongly negative in valence and therefore stressed, at a score of 9.13 with confidence 0.91, failing to separate that speaker's calm from stressed voice. This is the same out-of-distribution mechanism as the Sinhala case, but it surfaces with high rather than low confidence, so the defer-to-HRV gate does not trigger. The deployed product mitigates such cases through the faint-recording guard, the within-speaker verdict, and physiological deferral, and the residual risk is discussed in Chapter 7.

6.4 Integration results

Cross-modal integration was validated through the Layer 4 reconciliation path using stored and mock HRV readings, which drive the identical code for a live Component B reading would. In the end-to-end verification above, the voice and HRV trends agreed with an agreement value of 0.842, and the confidence gate behaved as specified, deferring to HRV when voice confidence fell below the 0.4 floor. The Component B client and its schema mapping are unit-

tested. A live end-to-end run against a running Component B with a streaming wearable remains outstanding and is scheduled as integration work, while the cross-modal logic itself is complete and exercised.

6.5 Summary of findings

The findings form one coherent account. Selecting frozen emotion embeddings by in-domain acted agreement is inverted on real stress (RQ1). The cause is a valence and arousal split in which valence transfers, but arousal does not, replicated at scale (RQ2). The split is specific to emotion pretraining rather than model size (RQ3). A valence-primary rule that follows from the finding improves separation on every checkpoint (RQ4). The resulting bilingual detector reaches 91.7 percent leave-one-out accuracy, strong for English and promising but under-determined for out-of-distribution Sinhala (RQ5).

7 DISCUSSION

7.1 Interpretation of findings

The five results tell a single story. A frozen emotion encoder carries the valence of real stress reliably but its arousal only weakly, and this asymmetry is what turns in-domain acted agreement, which rewards arousal agreement on expressive speech, into an inverted guide for real deployment. Selecting on it favours checkpoints tuned to the acted high-arousal manifold, which generalize worst to quiet real stress, whereas the checkpoint trained on naturalistic conversation, weak in-domain, generalizes best. Once stress is read from valence, every checkpoint improves and the ordering stops being harmful.

Arousal in speech rides mostly on prosodic energy and pitch dynamics, the very cues an acted delivery exaggerates and a quiet, internalized speaker suppresses, whereas valence rides on more distribution-stable tonal qualities that emotion pretraining captures well. The matched-size probe sharpens the point: the asymmetry belongs to emotion pretraining specifically, since a full-stack speech encoder shows the opposite bias. The lesson is not to always score on valence, but to know which affective axis an encoder transfers to the deployment distribution, and to score on that.

The result reframes from a routine engineering decision as a scientific question. Choosing a frozen encoder is usually treated as a matter of picking up the best-scoring checkpoint on a convenient benchmark, but the finding here shows that the benchmark can point in exactly the wrong direction for the deployment that matters. The useful habit is therefore not to trust the leaderboard but to ask, for the specific distribution the system will face, which affective axis the encoder preserves, and to build the scoring rule around that axis.

7.2 Comparison with literature baselines and existing solutions

The result refines rather than contradicts the surrounding literature. It is consistent with the established observation that acted and naturalistic emotional speech differ in distribution and recoverability [16], but adds that the transfer failure is asymmetric and axis specific. It aligns with spontaneous-speech findings such as EMOVOME, which report that spontaneous speech trails acted corpora [18], while explaining the mechanism behind that gap for stress. It supports the conclusion of the fundamental-frequency meta-analysis that voice alone struggles to separate stress from other aroused states and that complementary physiological measures are needed [8], and operationalises that conclusion as a confidence-gated cross-modal design. Against the convention of selecting frozen emotion models by in-domain agreement [11], [14], [15], the finding is a direct correction: on real stress this criterion is inverted.

7.3 Research contribution

The research contribution is fourfold. First, it demonstrates that ranking frozen emotion embeddings by in-domain acted agreement is anti-correlated with real-voice stress accuracy, with a Spearman rank correlation of -0.60, reversing a common selection habit. Second, it explains the cause as a valence and arousal split and replicates it at scale on IEMOCAP with bootstrap significance. Third, it shows that the split is specific to emotion pretraining rather

than model size using a matched-size probe. Fourth, it derives and validates a valence-primary scoring rule that follows from the finding and widens separation on every checkpoint, with a Cohen's d of 1.71.

7.4 Product contribution

The product contribution is an engineered five-layer component that deploys the finding as a working system: a bilingual English and Sinhala voice-stress detector reaching 91.7% leave-one-out accuracy, integrated with physiological HRV through a confidence-gated cross-modal framework, wrapped in a companion-guided mobile check-in that captures natural speech without a record control, and made robust by graceful fallbacks so the user experience never stalls. The design encodes the research lesson structurally, so the system's behaviour follows directly from what the evidence shows the voice can and cannot support.

7.5 Limitations and threats to validity

7.5.1 Statistical conclusion validity

The real-voice set is small, at 24 clips with roughly three Sinhala speakers. Leave-one-out thresholding removes fit-to-test bias but not small-sample variance, which is why per-language intervals are wide and why statistical inference is carried by the IEMOCAP replication rather than by the 24-clip p-values.

7.5.2 Construct and internal validity

Stress labels for the real set derive from the elicitation condition rather than a physiological gold standard, and the scores read acoustics rather than words. A notable internal threat surfaced in testing, where a calm out-of-distribution voice was scored stressed with high confidence. Because confidence is defined as the absolute value of valence, it presumes valence is trustworthy, an assumption that fails an out-of-distribution speaker whose valence is confidently wrong, so the defer-to-HRV gate does not trigger. This is mitigated by the faint-recording guard, the within-speaker verdict, and physiological deferral, but it remains a genuine residual risk.

7.5.3 External validity

Sinhala is out of distribution for the frozen encoder, and this cannot simply be trained away, as a controlled study showed that adding 26 Sinhala clips lowered held-out accuracy. Arousal magnitude is unreliable by design and is offloaded rather than repaired, and the IEMOCAP-trained checkpoint is used only as a reference and is excluded from the replication claim.

7.6 Implications

For practice, the primary implication is methodological: systems that reuse frozen emotion encoders for voice-based stress should first characterise which affective axis actually transfers to their deployment distribution, and select and score accordingly, rather than trusting in-domain agreement on acted benchmarks. For the wider platform, the implication is a clear division of labour, where the voice contributes a dependable valence-driven estimate, and the physiological channel contributes arousal magnitude. For low-resource languages, the

implication is that a frozen out-of-distribution encoder sets a ceiling that more in-language data alone will not lift, so meaningful gains require either genuinely diverse in-language data at scale or a lightly adapted encoder.

There is also a practical implication for how multimodal well-being systems should be built. Rather than treating each modality as a competing estimate of the same quantity and averaging them, it is more robust to assign each modality the part of the signal it can support, and to let a confidence gate decide when one should defer to the other. In this component, the voice supplies valence, and the physiological channel supplies arousal, and the gate makes the handover explicit, which is a division of labour that other combinations of behavioral and physiological sensing could adopt.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

On real, spontaneous stress, ranking frozen emotion models by their in-domain accuracy on acted speech is not merely uninformative but inverted. The reason is a valence and arousal split: these embeddings carry the valence of real stress but not its arousal, a pattern replicated on a large independent corpus and traced specifically to emotion pretraining. Reading stress from valence follows directly from the finding and widens the separation between stressed and calm speech on every model tested, giving a 91.7 percent leave-one-out detector deployed bilingually for English and Sinhala, with arousal deferred by design to a physiological channel. This thesis therefore delivers both a diagnosis and the design that follows from it, embodied in a working five-layer component within the wider stress-aware VR meditation system.

8.2 Achievement of objectives

The objectives of Section 1.5 were met as follows. The ambient assessment (1), the frozen-encoder model and axis-transfer ablation (2), the test of the selection criterion with large-sample replication (3), the derivation and validation of the valence-primary rule (4), and the change and cross-modal framework (5) were completed. The autoencoder anomaly detector (6) was designed, trained, and verified end-to-end. The bilingual delivery target (7) was met, with the shipped model exceeding the 75 percent threshold in both English and Sinhala on held-out real data. The one objective completed in part is the live cross-modal integration, which is implemented and unit-tested and was validated with stored and mock HRV, while a full end-to-end run against a live Component B remains outstanding.

8.3 Individual research and product contributions

The individual research contribution is the finding that frozen emotion embeddings transfer valence but not arousal to real stress, the demonstration that this inverts the conventional in-domain selection criterion, the large-sample replication on IEMOCAP, the matched-size evidence that the split is specific to emotion pretraining, and the valence-primary scoring rule that follows. The individual product contribution is the engineered five-layer, bilingual voice-stress component, with confidence-gated cross-modal validation, companion-guided capture, and graceful degradation, that operationalises the finding as a working system reaching 91.7 percent leave-one-out accuracy.

8.4 Future directions

Four directions follow from the limitations. First, grow the Sinhala speaker pool substantially, since the ceiling is speaker and distribution diversity rather than clip count. Second, calibrate the faint-recording and quality thresholds on live-session data so the confident single-speaker misread is caught, for example through an out-of-distribution term in the confidence estimate rather than valence magnitude alone. Third, test whether a lightly adapted rather than fully frozen encoder can recover arousal for internalized stress without losing the valence transfer

that carries the system today. Fourth, complete the live end-to-end integration with Component B, extend to cloud deployment, and add Tamil scoring at scale.

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GLOSSARY

Valence: the pleasant to unpleasant dimension of affect; the axis this component reads to score stress.

Arousal: the calm to activated dimension of affect; unreliable from the voice for internalized stress and deferred to the physiological channel.

Frozen encoder: a pretrained model used only as a fixed feature extractor, with no weights updated during training.

Concordance correlation coefficient (CCC): an agreement metric that penalizes both scale and location error.

Leave-one-out thresholding: setting the decision threshold on all clips but the one being scored, removing fit-to-test bias.

Prosody: the suprasegmental properties of speech such as pitch, energy, timing, and voice quality.

Valence-primary scoring: deriving the stress score from negative valence rather than from arousal.

Out-of-distribution: input unlike the encoder's pretraining data, for example Sinhala, where behaviors are less reliable.

APPENDICES

Appendix A Research instruments or extended results

This appendix provides extended per-checkpoint results that support Chapter 6. Table A1 reports the separation between stressed and calm speech under the conventional and the valence-primary rules, and Table A2 reports the valence and arousal recovery on real stressed clips.

Checkpoint	Separation (conventional)	Separation (valence-primary)
Fusion-MELD (shipped)	1.86	3.62
Fusion-V2	2.23	2.80
Fusion-IEMOCAP	1.19	1.94
Fusion-Acted	1.11	1.86
Fusion-Combined	1.18	1.61

Table A1: Separation between stressed and calm speech by scoring rule, per checkpoint.

Checkpoint	Valence correct (%)	Arousals correct (%)
Fusion-Acted	82	12
Fusion-Augmented	94	29
Fusion-Combined	100	29
Fusion-IEMOCAP	100	53
Fusion-MELD (shipped)	100	18
Fusion-V2	94	29

Table A2: Valence and arousal recovery on real stressed clips, per checkpoint.

Table A3 consolidates the headline per-checkpoint numbers that underpin Chapter 6: the in-domain acted agreement used by the conventional selection criterion, the real-voice accuracy, and the stressed-versus-calm separation under the conventional and the valence-primary scoring rules. The inversion between the first two columns and the consistent improvement in the last two are the quantitative core of the thesis.

Checkpoint	In-domain CCC	Real-voice acc.	Separation (legacy)	Separation (valence-primary)
Fusion-MELD (shipped)	0.35	0.917	1.86	3.62
Fusion-V2	0.79	0.750	2.23	2.80
Fusion-Combined	0.77	0.625	1.18	1.61
Fusion-IEMOCAP	0.65	0.583	1.19	1.94
Fusion-Acted	0.81	0.375	1.11	1.86

Table A3: Consolidated per-checkpoint results: in-domain agreement, real-voice accuracy, and separation by scoring rule.

Table A4 lists the principal configuration constants that govern the layers, all of which are configuration-driven rather than hard-coded in logic, so they can be recalibrated without changing the code.

Constant	Value	Role
Ambient good threshold	-45 dBFS	Room reads as quiet at or below this noise floor
Ambient too-noisy threshold	-30 dBFS	Session blocked above this noise floor
Loudness target (RMS)	0.05	Loudness normalisation target
Faint-input threshold (RMS)	0.02	Below this, confidence is reduced
Confidence gate	0.4	Below this voice confidence, Layer 4 defers to HRV
Stressed threshold	2.0	Score above this is flagged stressed (leave-one-out validated)
Anomaly threshold	mean + 3 standard deviations	Reconstruction error above this flags a session
Per-user threshold switch	5 sessions	After this many sessions, a per-user threshold is used

Table A4: Principal configuration constants governing the five layers.

Appendix B Academic integrity and AI use disclosure

This disclosure is provided in accordance with the module's academic integrity and responsible-AI policy. The research reported in this thesis, namely the problem framing, experimental design, the models and checkpoints, the data handling, the empirical results, and the findings and their interpretation, is the author's own original work.

Generative-AI assistance was used in a supporting capacity only, for language editing and proof-reading, for formatting the document into the provided template, and for scaffolding routine, non-research code during development. It was not used to design experiments, to produce or alter any result, or to generate scientific claims of the thesis.

All quantitative results, tables, figures, and claims were verified by the author against the primary sources, namely the author's research outputs, the ablation records, and the component's source code and tests, and all references were confirmed against their originals. The author takes full responsibility for the accuracy, originality, and integrity of the entire thesis.